What is a Virtual Machine?

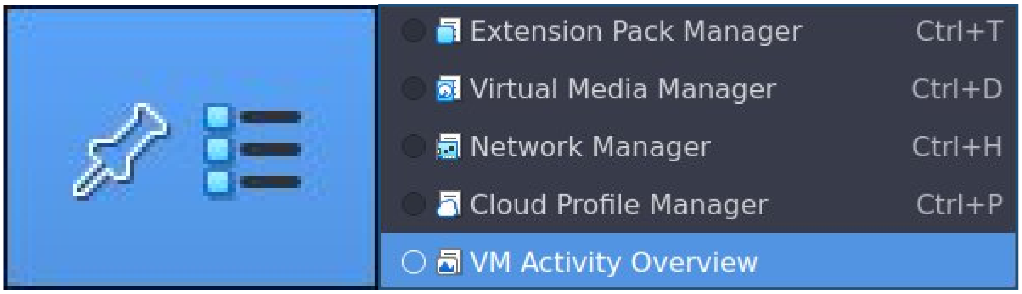
The way I like to explain it is “running another computer on your main computer”. There’s definitely more to it, but for now we can work with that. Since setting up a physical home lab with its own network equipment and multiple devices can be costly, I highly recommend virtualization. Machines can be created/destroyed very easily and you only need 1 physical device.

There are a few different options for running virtual machines. VmWare is popular, Virtualbox is my favorite, and Linux has it’s own solution Qemu which comes with most Linux distros. At some point I started using VirtualBox and it’s been my preferred software since. I also recommend looking into VmWare’s free version since it is so popular in enterprise settings. Everything we’ll be doing here will translate across the different software, but may be slightly different depending on what you use.

VirtualBox Primer

VirtualBox is backed by Oracle and I personally find it super easy to use. When you launch VirtualBox you’re greeted with the welcome screen shown below. I have a few machines already setup and you can see those listed on the left hand side under **Tools**. Let’s take a quick tour before we setup up a virtual machine.

Tools

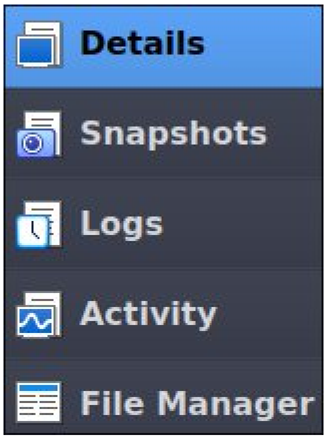


If you click the icon next to Tools (the three stacked lines) you'll open a dialog box with VirtualBox's different tools. You can also get to this list by going to File>Tools. Lets go over the different options briefly.

- **Extension Pack Manager**
  - This section lets you manager the different extension packs for VirtualBox. There is an official package for this provided on the VirtualBox website. *Note: When installing an extension pack make sure the version matches your VirtualBox version.*
- **Virtual Media Manager**
  - Here you can manage the storage devices in VirtualBox. You can create, add, and remove virtual hard disks here. You can also manage optical disks (such as iso files) and virtual Floppy Disks.
- **Network Manager**
  - In this section you can create and manage the different networks used by VirtualBox machines. Most important this is where you can manage the Host-Only Networks. Host-Only uses VirtualBox for routing which lets the different virtual machines talk to each other but not reach the internet.
- **Cloud Profile Manager**
  - This section lets you create a cloud profile which can be further configured under Network Manager>Cloud Networks.
- **VM Activity Overview**
  - Here you can get a read out of how many resources VirtualBox and the virtual machines are using in real-time.

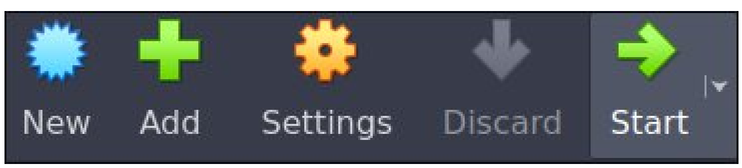
VM Tools

Just like with the tools above, you can access the virtual machine's different tools by clicking the icon next to the virtual machine name. These are the different options:



- **Details**
  - View the machine's specs
- **Snapshots**
  - Create, delete, or load machine snapshots
- **Logs**
  - View machine logs (great for troubleshooting)
- **Activity**
  - View resources being used in real-time
- **File Manager**
  - Manage files in the virtual machine and transfer files between the host and VM

Creating a VM

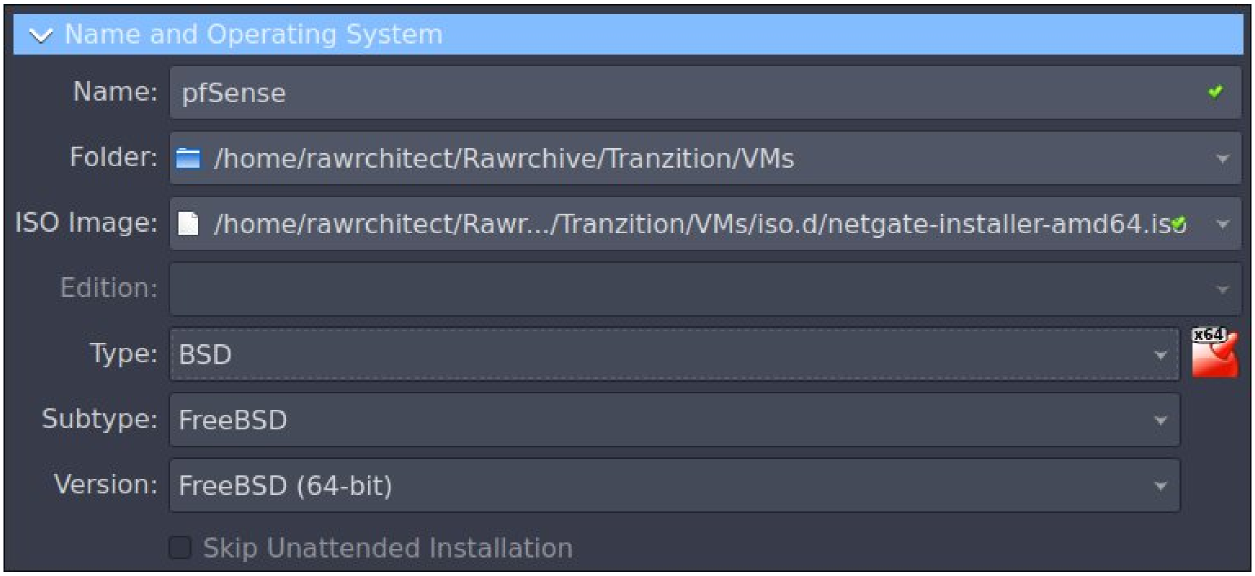


Now we can get to the fun part. For this example we'll be creating a virtual machine to run pfSense, a firewall that can be also used for routing and more. From the Weclome screen you can simply click New from the toolbar. If you've been browsing around VirtualBox and need to find the option you can select the Tools icon and then Welcome from the menu. Another way is to click Details from in a virtual machine's tools. This will bring back the necessary toolbar at the top.

*Note: Remember to select **New** when **creating** a machine. The **Add** option is used when adding an **existing** VM to VirtualBox.*

Name and OS

When creating a virtual machine we need to specify the operating system, any iso we need to install the OS, and set the amount of resources the VM can use. First is the OS section.



Here you can name the machine (pfSense in this case), and pick the location (Folder) to store the VM. After that is the ISO Image. Here you can select the file you want to use to install the OS. For pfSense that would be *netgate-installer-amd64.iso*.

When you select your iso file the Type, Subtype, and Version fields will automatically populate. This can be adjusted to fit your operating system if necessary. Be sure to look at the different options to become familiar with how these fields work. In this example Type and Version were set to “Other”. However, a quick search shows that pfSense uses BSD as it’s kernel base so selecting BSD and accepting the other defaults is what we’ll do here.

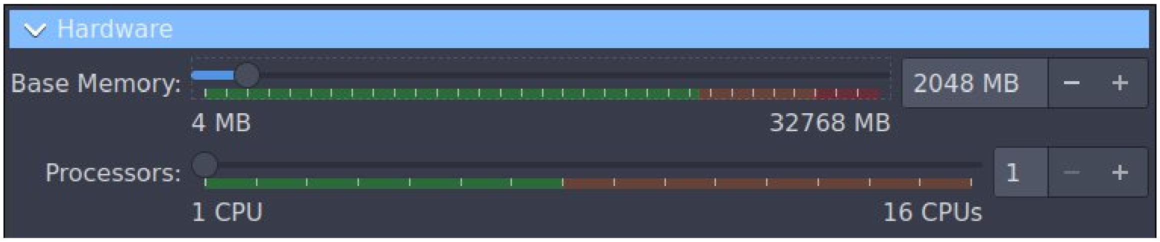
There’s also a check box here that we can select to skip the Unattended Installation. In this example the option is unavailable since Unattended Installation isn’t supported.

Unattended Install

The section after the OS section allows you to setup a user and password for an Unattended Install. This can be useful but not all operating systems allow for it. I like to skip this by default when creating VMs. You can also attach the Guest Additions ISO here. Guest Additions add extra functionality to the virtual machines (most notably enabling you to scale the screen to your native resolution). The Guest Additions can be installed on the VM at any time.

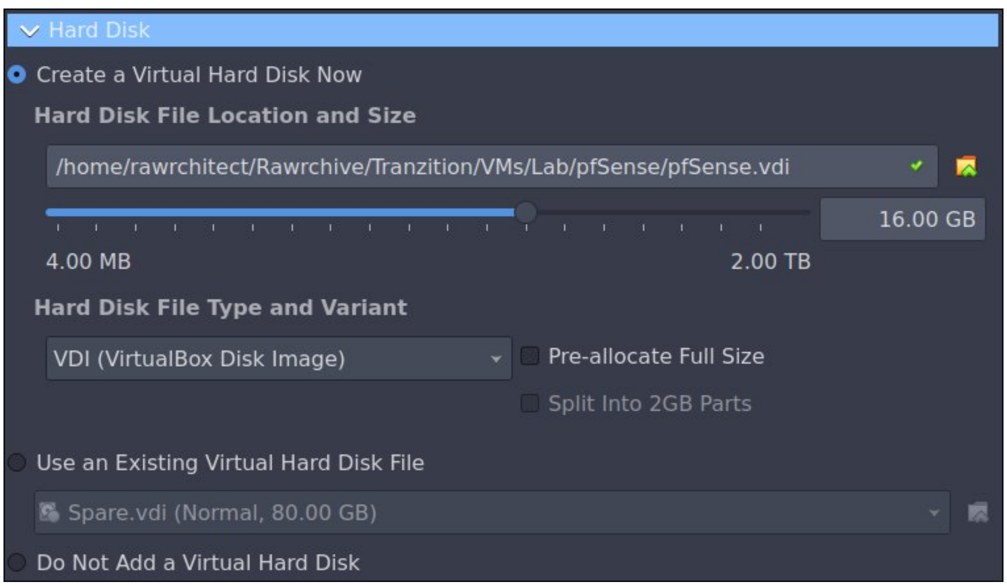
Hardware

When configuring the hardware options, it's a good idea to do a quick search on what resources you might need for each operating system. VirtualBox will set these with a default value, but those can be changed at any time. It's also important to take into account your system's resources and adjust accordingly. A lot of operating systems can be run with very minimal VM specs.



For our base memory the default was 1024 MB. This is fine and won't impact performance much. I have the resources to bump up the memory up a little more so I did. The default for Processors is 1 CPU and that will be more than enough for this machine.

Hard Disk



The last step is to create or add a Virtual Hard Disk. VirtualBox can use most virtual HD formats but the VirtualBox specific type is VDI. By default the total disk size you choose isn't allocated all at once. The virtual disk will grow as needed *up to* the amount you choose. If you want to go ahead and reserve all the disk space now you can select Pre-allocate Full Size.

The default for this machine was set to 10GB but to add some future proofing (just in case) I went with 16 GB. After configuring everything on this page simply click Finish at the bottom and you're all done! You still need to install the OS to actually use the virtual machine but we'll cover that in the *pfSense and Routing* walkthrough.

Machine Specs

Here are the specs of all my lab machines for reference. These specs can be adjusted to fit your resources but should be applicable on *most* systems.

pfSense

CPU: 1

RAM: 2048

Storage: 16GB

Windows Server

CPU: 1

RAM: 4096

Storage: 100GB

Workstation

CPU: 1

RAM: 2048

Storage: 80GB

Kali Machine

CPU: 2

RAM: 4096

Storage: 120GB